Report SAE S1.01

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Summary :

I. Introduction..... 2
II. Functionality..... 3
a) Capacity to answer questions..... 3
b) The no-answer case..... 3
c) Capacity of learning a new answer..... 3
d) Capacity of answering by taking the context in account..... 4
e) The “Thésaurus”..... 4
III. Problems encountered..... 4
IV. Comparison of the asymptotic computational complexity in the worst case between two approaches..... 5
a) The “naive” approach..... 5
b) Our approach using index..... 5
c) Acceleration factor..... 6

I. Introduction

The main goal of the project is to make a chatbot with Java that can quickly and rightly answer general knowledge questions in French. You can find an example of a dialogue between an user and the chatbot (bold) below :

> J'attends tes questions de culture générale.
> Qui a inventé le téléphone ?
> Le téléphone a été inventé par Alexander Graham Bell.
> Où ?
> Le téléphone a été inventé aux États-Unis.
> Combien y a-t-il de pays dans le monde ?
> Je ne sais pas.
> Je vais te l'apprendre.
> Je t'écoute.
> Il y a 195 pays dans le monde.
> Très bien, c'est noté.
> Combien y a-t-il de pays dans le monde ?
> Il y a 195 pays dans le monde.

Through this dialogue, we can observe that the chatbot gives a good answer. It also takes the context into account. It understand that “Où ?” is about the question on the telephone. Our chatbot is honest, it shows its ignorance instead of answering a question wrong. Finally, it is capable of saving information given by the user to use it later.

This report will present our work to reach this result, and analyse the technical efficacity of our approach compared to a naive approach.

II. Functionality

a) Capacity to answer questions

The chatbot can answer questions asked by the user by spotting the theme of the question using functions that isolate thematic words present in the question. Then, it spots the answers in his database that include the same thematic words. To choose an answer among the candidates, the chatbot isolates the grammatical form of the question and tries to make the isolated words correspond to those present in his database to finally link the question it was asked to an answer. If there are multiple matching answers to the question, the one that will appear will be randomly chosen.

b) The no-answer case

In cases where the answer to the question is not included in the chatbot's database, it will directly display the following message : "Je ne sais pas" and will not try to give a wrong answer.

c) Capacity of learning a new answer

By returning to the case where the chatbot displays "Je ne sais pas", we implemented a functionality that lets him learn the answer to the question he did not answer. By entering the following sentence : "Je vais te l'apprendre.", it will save the answer that the user will give. After restarting the chatbot, the newly given answer will still be present in its database. Moreover, by saving the new answer, it will enrich the grammatical form table with the new form and will improve its answering capacities.

Here is an example :

> J'attends tes questions de culture générale.
> De quel pays Paris est-elle la capitale ?
> Le pays dont la capitale est Paris est la France.
> De quel pays Nairobi est-elle la capitale ?
> Le pays dont la capitale est Nairobi est le Kenya.
> Nairobi est la capitale de quel pays ?
> Je ne sais pas.
> Paris est la capitale de quel pays ?
> Je ne sais pas.
> Je vais te l'apprendre.
> Je t'écoute.
> Paris est la capitale de la France.
> Très bien, c'est noté.
> Paris est la capitale de quel pays ?
> Paris est la capitale de la France.
> Nairobi est la capitale de quel pays ?
> Nairobi est la capitale du Kenya.

d) Capacity of answering by taking the context in account

The chatbot has the capacity of answering a question without thematic words (like “Où ?”, “Quand”, etc…) by taking the context of the last question that includes a theme into account. Thus, it is possible for the user to ask multiple questions on the same theme without needing to reenter the full question again.

e) The “Thésaurus”

The “thésaurus” is used to make the chatbot more intelligent by allowing him to understand the words’ meaning rather than just their exact writing. It is like a conversion dictionary that transforms a word’s variants (like the French feminine versions), synonyms (like “named” to “called”) and even spelling mistakes into an unique reference word. Thanks to the “thésaurus”, the chatbot can link a question to an answer although the words used in them are not identical.

III. Problems encountered

We encountered a problem with the function découpeEnMots that cuts a String variable in a String vector with each object being a word of the previous variable. This function is used a lot in the construction of the index. The problem was with the character used to cut the String. Some characters like **“-”** and **“ ’ ”** were not present in the function although they were used in the answer database. We then added them to the function and there were no more cutting problems.

IV. Comparison of the asymptotic computational complexity in the worst case between two approaches

a) The "naive" approach

Without the use of index, it is needed to browse through all the answers in the database, their thematic words and the thematic words of the questions, and compare them with the user's question.

We set :

- *nr* : number of answers
- *nmr* : average number of thematic words in an answer
- *nmq* : average number of thematic words in a question

The calculation is : **CoûtNaïf = *nr* x *nmr* x *nmq* =** 100,000 **x** 5 **x** 3 **=** 1,500,000 comparisons

The "naive" approach will make **1,500,000** comparisons in the worst case.

b) Our approach using index

With the use of index, it is not needed to browse through all the answers, their thematic words and the thematic words of the questions. Each distinct thematic word has in the index the answer's number in which it appears.

We set :

- *nmv* : number of distinct thematic words in all the answers
- *nmr* : average number of thematic words in an answer
- *nmq* : average number of thematic words in a question
- *nrpm* **= (***nr* **x** *nmr***) /** *nmv* : average number of answers that include a thematic word

Calculation :

CoûtRecherche = *nmq* **x log$_2$(***nmv***) =** 3 **x log$_2$(**10 000**) =** 3 x ~13,3 **=** 40 operations

Coût1ereFusion (Coût1) = *nrpm* **+** *nrpm* **=** 2 **x** 50 **=** 100 operations

Coût2emeFusion (Coût2) = (2 **x** *nrpm***) +** *nrpm* **=** 100 **+** 50 **=** 150 operations

CoûtMaxOccurence = *nmq* **x** *nrpm* = 3 **x** 50 = 150 operations

CoûtListes = **Coût1** + **Coût2** + **CoûtMaxOccurence** = 100 + 150 + 150 = 400

CoûtFinal = **CoûtListes** + **CoûtRecherche** = 400 + 40 = 440 comparisons

Our approach using index will make 440 comparisons in the worst case.

c) Acceleration factor

Facteur = **CoûtNaïve** / **CoûtIndex** = 1,500,000 / 440 = 3,409

The approach using index is approximately **3,400** times faster than the naive approach.